

Presenter Reference Notes – September 2026

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Slide 1: NDNSF: A Data-Centric Service Framework for

Secure, Collaborative, and Dynamic Networked Applications Department of Computer Science University of Memphis Advisor: Dr. Lan Wang PhD Proposal Defense September 2026

Slide 2: Service Discovery: Several Candidates, One Executor

The User knows the required service. Reachability, capability and willingness may change. Providers evaluate valid Requests and return positive or negative ACK Data. The User selects one eligible positive-ACK Provider. Candidate Providers A, B, C User selects A A executes the service Initial NDNSF: discover alternatives and select one Provider per invocation, not multiple collaborating roles. Arrows show the selection process.

Slide 3: From Invocation to Collaboration: Two Research Scopes

Foundation: evaluate discovery and authorization The initial prototype selects one Provider. Discovery confidentiality, service permissions and management costs matter even without collaboration. Extension: coordinate complementary roles Some tasks need several Providers with different responsibilities. Separate calls do not define a shared role assignment or enforce dependencies between them. Evaluate the invocation foundation; develop and validate the collaboration extension. Existing invocation results do not establish collaboration correctness.

Slide 4: NDN Primitives and Service-Level Decisions

NDN building block Remaining application responsibility Named Data and signatures [1] Names support identity and forwarding; a Trust Schema [16] constrains signing keys Interest retrieval and caching [1,26] Check task identity, dependencies and deadlines; caches do not guarantee storage Dataset synchronization / SVS [2,3] Discover names; define service-message transitions NAC [4] / enhanced NAC-ABE [19] Encrypt Content for read access; check invocation permission separately NAC: Name-based Access Control; ABE: attribute-based encryption; SVS: State Vector Sync. NDN supplies data-centric communication. NDNSF defines how validated service messages authorize and advance an invocation.

Slide 5: Application Needs Determine the Research Scope

Application need Framework question Find a camera covering an area, or compute capacity for a model Who is permitted, reachable and willing now? Analyze images from several cameras, or execute dependent model fragments Which Provider performs each role, and which inputs may it use? Obtain useful output before a deadline, despite loss or outages What completes, what fails, and what does recovery cost? These needs lead to RQ1–RQ3. Invocation [9,10] and command authorization [12] provide precedents. This is a bounded framework scope, not a claim to cover every application requirement.

Slide 6: Three Research Questions

RQ1: How can NDNSF discover willing Providers, keep descriptors confidential from unauthorized readers, and check service permissions before execution, and at what cost? RQ2: How can single-Provider invocation be extended to roles and named dependencies across Providers, whether Data comes from producers or caches? RQ3: When do these mechanisms improve completion, latency, or execution efficiency, and when do their costs outweigh the benefits? RQ1 evaluates the invocation foundation; RQ2 studies its collaboration extension; RQ3 measures both scopes separately.

Slide 7: Related Work and the Remaining Questions

Prior work What the comparison must acknowledge NFN [6], NFaaS [7] Named computation and function execution/placement RICE [9], NSC [10] Invocation mechanisms; NSC checks signed calls using a Trust Schema DNMP [12] Trust Schema constrains roles and measurement commands CaSCON [15] Service-composition research; requires a feature-level comparison before claiming a new composition mechanism gRPC [17,27] Complete configurations can include discovery/resolver integration, not only fixed addresses Collaboration research: make runtime assignments constrain intermediate Data use. Discovery and authorization are evaluated separately.

Slide 8: Research Contributions and Current Evidence

Intended contribution Current evidence boundary Discovery / authorization design and tradeoffs Invocation implementation and bounded experiments exist; full alternative lifecycle-cost comparison remains open ACK-derived plan and local input checks Design and implementation work; direct/cache and failure validation remains incomplete Cross-case empirical findings Existing mobility/work-efficiency and small-model results support limited configurations, not all new collaboration claims Proposal status, September 2026: implemented, locally tested, measured, and fully qualified are not interchangeable. [28]

Slide 9: Participants and Trust Bootstrap

User: requests a service and selects a Provider; the extension also assigns collaboration roles. Provider: offers and executes a service. A node may play both roles. Controller: administers identities and service permissions approved by the organization. Participants use provisioned trust anchors and validation rules for certificates and Controller material. Unknown runtime candidates are not unregistered authorization identities. The User discovers current participation; the Controller does not choose this task's Providers.

Slide 10: Invocation Overview: Request a Timely Aerial Image

An operator requests an aerial image. Authorized camera Providers A and B consider the task. User Request Data Candidates ACK Data User Selection Data Selected Provider Response Data A returns a positive ACK; B is busy and returns a negative ACK. The User selects A; A checks Selection before capturing the image. Arrows show application order. Data is retrieved with Interests; neither an ACK nor a network delivery alone authorizes execution.

Slide 11: Single-Provider Invocation: Four Named Data Types

Data Logical purpose Request Describe required service and participation constraints; include protected discovery information and RequestNonce ACK Report positive or negative participation; positive ACK

supplies selection information and ProviderChallenge Selection Assign selected execution; provide input/references and protected request-scoped keys; constructed using ProviderChallenge Response Return status, results or result references; the User still validates them before use Sync helps discover names; Interests retrieve Data. The Provider checks Selection before execution; a positive ACK alone does not start the task.

Slide 12: Request: Discovery Descriptor; Selection: Task Input

Discovery descriptor Needed to decide whether to participate: area, deadline, sensor/model and resource requirements. Application input Needed to execute the selected task: images, detailed processing parameters, or protected input references. Input and results use request-scoped symmetric keys. Selection delivers key material encrypted for the selected Provider; the requesting User retains the keys. The boundary follows necessity. A location needed to assess coverage belongs in the protected descriptor; an image needed only for processing can wait.

Slide 13: Three Security Requirements; One Additional Benefit

1. Keep discovery descriptors confidential from unauthorized readers before selecting a Provider.
2. Authenticate participants, verify service permissions, and reject replayed invocation messages.
3. Grant and withdraw service permissions during operation. Additional benefit: aggregate service permissions in a decryption key (DKEY) while retaining identity-signing credentials. This is a management property, not another security guarantee.

Slide 14: Reason 1: Confidential Discovery Before Selection

The User knows the service, but not yet the currently willing Provider identities. An observation area and deadline can reveal intended activity even without the full application input. The User encrypts the descriptor using ABE and the service attribute, without listing Provider identities. [19] Confidential from parties lacking the required decryption capability. Authorized but unselected Providers can read it; names and traffic patterns remain visible.

Slide 15: Reason 2: Reuse Service-Permission Management

ABE was introduced to keep discovery descriptors non-public. [19] NDNSF reuses the attribute authority to manage Provider and User service permissions. [28] RequestNonce and ProviderChallenge checks use these permissions during invocation. ABE discovery plus role-based invocation is also possible. NDNSF avoids adding role signing credentials solely for service membership; signature validation and request-state checks remain necessary. User DKEYs and challenge processing still add cost.

Slide 16: How NDNSF Checks Service Permissions

1. Provider permission: recover RequestNonce from ABE-protected Request content and return it in ACK Data. The User compares it with RequestNonce stored for this request.
2. User permission: recover ProviderChallenge from ABE-protected ACK content and construct Selection Data. The Provider validates Selection using its stored ProviderChallenge. Also check current service permissions. These checks show access to protected values, not exclusive key possession. RequestNonce is not the Interest Nonce.

Slide 17: Identity and Replay Checks Have Different Jobs

Identity authentication Validate Data signatures and certificate chains under trust rules. An accepted signing identity alone does not establish service permission. Replay protection Match messages to an active request and its state. Record whether ProviderChallenge has already been used to accept Selection for execution; duplicate Selection must not start another execution. Retain or safely invalidate request state, including across restart. Encryption or a challenge value alone does not prevent replay.

Slide 18: Additional Benefit: Aggregate Service Permissions

Within one Controller domain, combine several service permissions in one entity's DKEY; retain identity-signing credentials. [19,28] DNMP represents command authority through role keys and schema rules. One role can permit several commands. [12] Where equivalent permissions require distinct role credentials, NDNSF avoids maintaining those extra signing credentials. Fewer independently managed credentials does not imply fixed DKEY size or lower total cost. Policy size, key distribution and updates still matter.

Slide 19: DNMP: Trust-Schema Command Authorization

Distributed Network Measurement Protocol (DNMP) [12] authorizes measurement commands: 1. The Network Key anchors role keys, device keys, and the schema. 2. The client uses target, probe type, arguments, and user role to find an allowed signing key; among matching keys, it chooses the least privileged. 3. The receiving Network Observer Daemon validates the publication under the schema before dispatching the probe. DNMP also uses publication lifetimes and retained state against replay, assuming loosely synchronized clocks. These are concrete command-authorization mechanisms, not just identity checks.

Slide 20: DNMP-inspired Authorization versus NDNSF

Design Execution authorization Discovery confidentiality DNMP-inspired [12] Check role-constrained signatures and NDNSF invocation state Pair with service-group encryption or ABE NDNSF ABE-backed Check signatures, RequestNonce, ProviderChallenge, current permissions and request state User encrypts descriptor content with service-level ABE Proposed comparison: retain NDNSF discovery and Selection, with input/result protection and revocation checks in both designs. Trust Schema [16] alone does not encrypt Content; adapting DNMP authorization is not an original DNMP feature. No measured cost advantage is claimed.

Slide 21: When Another Authorization Design May Be Better

Known Providers and simple rules: signed invocation plus recipient encryption may be sufficient. A small, stable receiver group: ordinary group-key management may be attractive. Rich command-argument restrictions: DNMP-style rules make those constraints explicit. [12] Many changing entitlements: measure both policy/key updates and credential/rule updates before claiming a management advantage. Compare complete combinations of authorization and encryption. ABE-backed authorization is one design choice, not a universally superior solution.

Slide 22: Runtime Permission Grants and Withdrawals

Both operations update Controller-signed permission status. [28] Ordinary grant: update the target's complete policy; it retrieves a replacement DKEY. Public parameters stay unchanged. Withdrawal:

rotate the Controller's ABE master key and public parameters; remaining members obtain DKEYs for their remaining permissions. The master key stays with the Controller. Rotation is Controller-wide. Participants must retrieve and validate updates before applying them; disconnected nodes cannot do so immediately.

Slide 23: Discovering Updates Does Not Make Revocation Instantaneous

1. Revised design: the Controller publishes new public parameters as signed Data; Sync announces availability. 2. Participants retrieve and validate parameters and permission status, then refresh DKEYs as required. 3. Grants also require status discovery, even when public parameters do not change. Local status retrieval and rotation exist; dedicated Sync announcements still need end-to-end verification. Measure propagation, rejection delay and rekey traffic. Neither NDNFS nor a certificate-revocation alternative can recall disclosed keys or plaintext.

Slide 24: Threat Model: What Is Trusted?

Trusted: Controller authorization and the User's planning decision. Untrusted delivery: an object may arrive through a producer, forwarder, cache, or repository and still requires validation. Adversarial cases to test: wrong identities, permissions, challenges, request state, roles, dependencies, or execution identities. Not established by signatures/ABE: honest resource reports, correct inference, truthful images, or resistance to authorized key sharing. No Byzantine-coordinator consensus claim; no traffic-analysis or instantaneous-revocation guarantee.

Slide 25: RQ2 Extension: From Positive ACKs to a Role Plan

1. Application supplies roles, dependency template, constraints and deadline; the template may be handwritten. 2. The User closes the ACK window and freezes the eligible, validated positive ACKs. 3. The User applies a replaceable planning policy and signs the resulting role assignments and dependencies. 4. Each Provider validates the plan, derives permitted inputs and rechecks its resources before starting. The snapshot freezes observed candidates for this decision, not every Provider in the network or their future availability.

Slide 26: A Plan Is More Than a Request ID

Existing request ID Proposed task-specific plan rules Identifies the request; requires a validated message Assigns exact Providers to collection and processing roles Does not alone specify predecessors Declares which producer role may supply each named dependency Does not alone describe key scope Associates permitted objects, execution identity, and authorization/key scope Example: within the same request, a service-authorized UAV that was not assigned "view 1" must not satisfy the view-1 dependency.

Slide 27: Plan-to-Data Enforcement: Check Before Advancing

1. Validate the Data signature using the appropriate trust rules. 2. Check active request, plan, execution identity and deadline. 3. Verify that the signer is the assigned predecessor Provider. 4. Check that the object matches the declared edge, permitted name, data format and key scope. 5. Require complete validated input before the dependent computation starts. Apply the same rules to direct, cached and

repository-fetched Data. This is a proposed acceptance contract; full end-to-end qualification remains required.

Slide 28: Collaboration Recovery Needs Explicit State

Event Required behavior Late ACK Do not silently change the committed plan Duplicate Selection Check whether execution was already authorized; do not repeat a side effect Incomplete dependency Wait within policy/deadline, recover, or fail; do not execute on partial data Provider replacement Authorize a new assignment; reject old execution actions and specify permitted result reuse Cancellation Prevent late messages from reauthorizing cancelled work Network re-retrieval is not task re-execution. Compensation is not an atomic transaction or exactly-once physical action.

Slide 29: NDN-SF-UAV: Application-Level Validation

Unmanned aerial vehicle (UAV) application workflows exercise several framework mechanisms. [28] Application function NDN-SF validation scope Control, telemetry, mission status Provider/request association, permissions, operation lifecycle Live video and recordings Stream lifecycle, protected delivery, named-object retrieval Detection and multi-view tasks Service provision/consumption, role assignments, dependencies Coverage is broader than one collaboration case. Each claim still needs evidence from the actual NDN-SF path; flight safety and image-analysis accuracy need separate validation.

Slide 30: UAV Collaboration Example: Capture, Analyze and Return

An operator requests ground-area views from nearby UAVs, followed by analysis. 1. After Selection, assigned UAVs capture views and sign encrypted ViewImage Data under their authorized namespaces. 2. The compute Provider retrieves images over assigned edges and publishes Annotation Data. 3. Its encrypted result gives the requester original image names and permitted image keys. 4. The requester retrieves the original images and displays the annotations. Original image Data remains under its producer's name. GPS alone does not establish camera visibility.

Slide 31: UAV Evaluation: Completion and Correct Input Use

A complete authorized collection-processing-return task before the deadline. Correct handling of unavailable views, delayed images and a failed processing Provider. Reject wrong-role/edge inputs; accept cached images only when they satisfy the same task checks. MiniNDN [22] and PX4 SITL [23] results clearly distinguished from physical-flight evidence. Higher recognition/localization accuracy is a separate claim requiring ground truth and a single-view control. It is not assumed here.

Slide 32: NDN-SF-DI: Application-Level Validation

Application function NDN-SF validation scope Single-Provider inference Discovery, permissions, input and result delivery Model preparation and reuse Readiness, artifact references, named-object retrieval Incremental output and conversation Request state, cancellation, output lifecycle Multi-Provider inference Assignments, dependencies, recovery The application also tests the invocation foundation, without requiring a model split. Interfaces and focused tests exist; complete model/device/recovery qualification remains open. [28]

Slide 33: DI Collaboration Example: Named Computation Dependencies

Provider A layer group 1 Provider B layer group 2 Provider C output stage activation activation Pipeline: sequential layers/groups; tensor partitioning additionally needs supported collectives. ONNX [25] describes the graph; export alone does not implement distributed execution. Repo [24] stores artifacts; forwarding caches do not guarantee persistence. The framework contribution is checked runtime assignment and data exchange, not model partitioning itself.

Slide 34: DI State and Evidence Boundaries

Model weights, reusable artifacts, request activations and KV state have different lifetimes. With KV caching, the runtime reuses prior key/value (K/V) tensors and supplies next-step inputs; resident weights do not replace this state. [31] An in-process test does not establish correct execution across processes or devices. Freeze one supported model/partition first; tensor/hybrid and LLM streaming need their own evidence if claimed. Earlier small-model measurements do not validate the newer distributed inference and recovery paths. [28]

Slide 35: Application Coverage and Observable Outcomes

1. Name the application function and the production NDNSF API path it uses. 2. Specify the input, expected result, failure condition, and tested configuration. 3. Record evidence separately: implementation, focused test, multi-process test, measured outcome. UAV and DI cover discovery, authorization, data delivery, lifecycle, and collaboration. Local calls do not prove remote behavior. Representative workflows cover claimed mechanisms; optional application features need not become separate dissertation requirements.

Slide 36: RQ3: Define the Baselines Before Comparing

NSC [10]: named service calls; describe the tested executor, timeout, queue and recovery behavior. gRPC [17]: describe resolver, sequential/parallel policy, deadlines, retry and cancellation. Match Provider count, capacity, task, topology/traces, publication times and measurement windows. Report fixed-list and dynamically discovered configurations as different configurations. An incomplete baseline cannot establish a limitation of the full framework. Historical unmatched data remain diagnostics.

Slide 37: Mobility: Completion under Different Coverage Conditions

Ten seeds (62–71), four Providers/system, 2 m/s; mean seed completion rate. [28] AP radius NDNSF gRPC NSC 50 m 54.57% 53.80% 55.47% 100 m 97.97% 97.97% 98.33% 150 m 100.00% 100.00% 100.00% No overall NDNSF advantage is established. The next slide uses a separate holdout campaign (seeds 72–81), not a subset of these runs.

Slide 38: Mobility: Lower Tail Latency in a Defined Subset

Seeds 72–81; four Providers/system; 100 m, 2 m/s; 1,312 aligned switching requests. [28] Comparison Seed-p95 difference 95% CI NDNSF minus gRPC-SEQ-4 -597 ms [-992, -200] ms NDNSF minus NSC-4 -2618 ms [-2916, -2315] ms Completed: NDNSF 1,311; gRPC 1,297; NSC 1,296 (of 1,312 each). Mean

paired seed-p95 differences; each p95 uses that system's successful requests. Different failures limit the comparison; no general advantage is inferred.

Slide 39: Fewer Execution Starts than the Parallel Baseline

Six-seed, capacity-matched comparison; 1,800 requests per system. [28] Configuration Completed requests Execution starts NDNSF 1,798 / 1,800 1,800 gRPC-PAR-4 1,800 / 1,800 7,180 NSC-4 also completed 1,800/1,800, with 1.002 starts/request (rounded). No work-efficiency advantage over NSC is claimed. Fewer starts than gRPC-PAR-4 is not proof of fewer bytes, less CPU time or lower energy. Similar observed completion is not proven equivalence.

Slide 40: A Useful Negative Result: Splitting Can Be Slower

Earlier small Qwen-style ONNX model; ten successful runs per layout. [28] Layout Mean latency p95 Shared-backbone distributed layout 259.4 ms 283.9 ms Single Provider 186.8 ms 206.3 ms Do not split merely because several Providers exist. Communication/setup overhead is a candidate explanation, not a causal result established by this measurement. Retained historical diagnostic; raw-run/model provenance remains to be rechecked. Not evidence for the newer distributed design.

Slide 41: Remaining Experiments Follow the Claims

RQ Required evidence Outcome and cost RQ1 ABE-backed versus DNMP-inspired rules plus encryption; unauthorized call, replay, grant and withdrawal No unauthorized handler start; discovery delay, key/policy size, status delay and rekey traffic RQ2 One UAV and one DI graph; wrong role/edge; direct/cache parity; cancellation/replacement Correct task completion; validation, transfer and recovery cost RQ3 Matched workload and failure schedules; feasible single-device control; cold/warm state Completion plus latency, redundant work, bytes, uncertainty, and cases with no advantage Planned tests are not completed findings. Record all scheduled runs and failure boundaries.

Slide 42: A Bounded Path to Spring 2027

Internal target Required deliverable September 2026 Agree scope/RQs; freeze one UAV graph and one DI partition; resolve correctness blockers October–November 2026 Complete registered experiments; freeze evidence and write bounded RQ answers December 2026 Complete integrated advisor-review draft January 2027 Committee review and readiness decision; reduce scope or revise timing if core evidence is missing February 2027 Target defense, conditional on evidence and committee approval Committee draft: two weeks before defense. Spring 2027 application/candidacy: February 5; final ProQuest: March 26. Recheck before filing. [29,30]

Slide 43: Expected Research Outcomes

RQ1 answer: when ABE-backed authorization is preferable to DNMP-inspired role rules plus encryption, accounting for confidentiality, management and withdrawal costs. RQ2 answer: exactly which task-assignment mistakes are rejected and which recovery behavior is supported. RQ3 answer: measured benefits, negative results and deployment limits. Reproducible configurations and evidence linking each conclusion to its tested scope. The outcome is a tested service protocol and an evidence-backed account of when its mechanisms are worth their cost.

Slide 44: Discussion: Is This Scope Sufficient and Feasible?

1. Are the three research questions specific enough to support defensible answers? 2. Are the authorization alternatives complete and fairly configured? 3. Are one UAV graph and one supported DI partition sufficient to test the abstraction? 4. Which optional features should be excluded if they delay the core evidence? NDNF: evaluate runtime discovery and authorized invocation; extend the foundation to role-aware collaboration.

Slide 45: Backup: Discovery Configuration Is Not Framework Capability

Provider-transition configuration Completed NDNF, no pre-registered Provider addresses 357 / 357 Static-list NSC/gRPC configurations 0 / 357 Pre-registered baseline variants 357 / 357 This isolates the configured discovery boundary. Resolver/discovery integration can update candidates; it is not proof that gRPC or NSC cannot do so. [27,28]

Slide 46: Backup: Historical Loss Evidence Is Not a New Matched Test

Historical per-link loss NDNF completed requests 1% 599 / 600 3% 598 / 600 10% (best recorded run) 539 / 600 A selected run is not an aggregate estimate. Do not combine these numbers with another batch and label the result “matched.” [28] Per-link packet loss is not the probability of service failure: retrieval retries and the request deadline also affect completion.

Slide 47: References: NDN and Authorization

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Slide 48: References: Invocation and Composition

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Slide 49: References: Tools, Evidence and Degree Planning

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